

Walking On Water

When we get back to the traditional way of looking at it, we can see that we would need lots of memory to store intermediate results—at each point in the computation, the results would be stored somewhere, instead of being thrown away (and therefore dissipated as heat). The next step would generate more results, and so on, and the memory would soon fill up. For a really complex calculation, you would need tons and tons of RAM! So what we do is, we go forward a few steps, and reach a certain stage x . Then we take all the results we've accumulated, and use those in addition to the information x —and do some more calculations. We thus proceed a little bit further. Then we backtrack again, and...

To give a simple analogy, think of crossing a river using only a few stepping stones (which represent the limited memory). You place, say, five of them in the river, and walk to a point. Then you stand on the fifth stone, pick up the other four, and put them in front of you. You discard what was the fifth stone, then walk ahead, pick up the stones again, and...

Think of it thus:

$5 + 3 = 8$ (the first step; we don't throw away the knowledge that 5 plus 3 equals 8)

$6 + 1 = 7$ (the second step; we don't throw away the knowledge that 6 plus 1 equals 7)

Say we get to a step that involves adding 5 and 3 and 9, we don't do $5 + 3 + 9$; we use our prior knowledge and just do $8 + 9$.

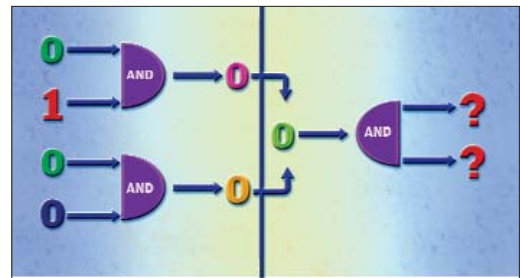
That's a crude (and strictly speaking, incorrect) way of looking at it, but it does reveal what the worth of "uncomputing" values is. The challenge lies in devising logic where uncomputed values can be put to use.

Logically Speaking

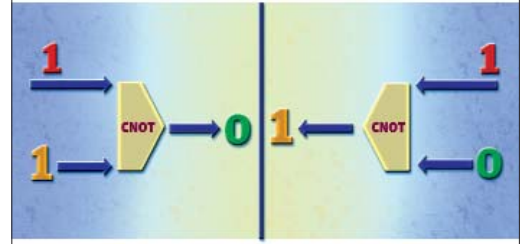
We've given you two fancy, fun ways of looking at how a reversible computation would work. But what would a reversible circuit actually look like?

We need to get back to logic—we need reversible logic gates. As a simple example, the AND gate is irreversible (like we said), but the NOT gate is reversible—you take the output and feed it back, and you get the original input.

Now think of a CNOT (Controlled Not) gate: it has two inputs A and B, and it flips B if and only if A is 1. In other words, it's a NOT gate with a control line. Then think of the Toffoli gate, which is a CCNOT (Controlled controlled NOT) gate: it has inputs A, B, and C, and it inverts C if



An AND gate. Different input sets give the same output. How can you possibly get back what you put in?



A CNOT gate. Reverse the operation, and you get back what you had. Of course, even the NOT gate is an example

and only if A and B are both 1. It turns out that the Toffoli gate is a "universal" reversible gate, in the sense that (in simple terms) any reversible computer can be built using them.

But that's only in theory. The real gates that a reversible computer would be based on are more complex, and space does not permit a discussion of them here.

Reality Check

We now know how reversible computing has the potential to reduce heat production, and to enable chips to shrink further. But is it just another fad? A toy technology, so to speak?

It simply is not. The general ideas of the technology date back to the 1970s! ITArchitect.com surmised two years ago that in 2013, adiabatic techniques will become more cost-effective than traditional methods for desktops. With performance per watt becoming more and more of a benchmark, the trend is clearly towards "green computing."

The trend is also towards quantum computing, which has been much in the press the last few years (refer our nameless article on quantum cryptography, *Digit* February 2006). And there is a close relationship between quantum and reversible computers: every quantum computer is a reversible computer. But before quantum computing arrives, problems in reversible computing need to be sorted out.

We'll give the last word to Dr Michael Frank, now a junior faculty member in the ECE department in the FAMU-FSU College of Engineering, and a Reversible Computing pioneer. Dr Frank said in an interview with *Digit* in 2004: "It is not an exaggeration to say that the entire long-term future of computing technology, and indeed of our entire civilisation, will depend on exploring the problems that reversible computing poses."

Essentially, the merger of physics and computer science is the new convergence we're headed towards. ■

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Adiabatic Circuits

As you might have gathered, the actual circuits that a reversible computer will use will be highly adiabatic. Now digital logic, of course, encodes bits using two voltage levels, and uses transistors to connect and disconnect nodes, and for the power supply, so that the voltage levels can be manipulated. Adiabatic logic uses the same general approach, but new rules must be followed.

❑ Never turn On a transistor when there is a voltage difference between its terminals

❑ Never turn Off a transistor when there is a current passing through it.

These are irreversible events, and therefore strict no-nos in the world of Reversible Computing.